

SEEMAB — CLASS 9 (FEDERAL BOARD)

Mathematics exam-18 (Version B) — Detailed Marking Report

Units 1, 2, 4, 5 and 7 • 38 marks, no choices • attempted and marked 15 September 2026 • independently deep-checked

32.5/38 TOTAL	85.5% PERCENTAGE	A GRADE	7/10 SECTION A	10.5/12 SECTION B	15/16 SECTION C
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The short version. Seemab scored 32.5 out of 38. Her written work was strong again at 25.5/28, and the long questions lost a single mark. The paper was built to re-test her past mistakes, and most of them held: the sign rules for sums and differences of cubes, absolute value, the minus sign on an inequality's lower bound, verifying both roots, naming inequality properties, and writing the collinear conclusion. What did not hold is the ordered-pair question ($a = 9$ again, the same slip as exam-16) and the habit of stopping one line short, which cost her the characteristic of $\log 0.0567$ and the final line of the square root.

A new pattern is worth attention: on two MCQs her working reached the right answer but the letter she wrote did not, and she skipped one MCQ entirely because her numbering went off track.

SECTION A — Multiple Choice (7 / 10)

Q	Topic	Hers	Key	What happened
1	$\sqrt{98} + \sqrt{8}$	B	B	Correct. But see the note below: she wrote that she used a calculator.
2	$\log 6 + \log 5$	A	A	Correct, with $\log(6 \times 5) = \log 30$ worked out.
3	$\log_3(1/27)$	none	C	Not answered. Her labels run 1, 2, 3, 4, 5, 7, 8, 9, 10, and her "3", "4" and "5" carry the working for paper Q4, Q5 and Q6. She lost count and skipped this one.
4	$27m^3 + 64n^3$	D	D	Correct, with full working. Sum-of-cubes sign: fixed (missed on exam-15 and exam-16).
5	$27 - 8y^3$	A	A	Correct, with full working.
6	$ -9 - -4 $	B	B	Correct: $9 - 4 = 5$. Absolute value: fixed (exam-17 Q7 read the bars as brackets).
7	$-4 \leq 2x - 2 \leq 6$	D	C	Working right, letter wrong. Her working ends $-1 \leq x \leq 4$, which is option C, but she wrote D. The exam-17 slip (dropping the minus) is fixed; this is a copying error.
8	Additive property	A	A	Correct. Her only reason: "because other options doesn't make sense".
9	$(2a - 1, b + 3) = (9, 5)$	A	D	Same mistake as exam-16 Q22. She substituted $a = 5$ and got $2(5) - 1 = 9$, then boxed "a: 9" and " $5 = b$ ", giving the value of each element instead of the letter.
10	Mid-point	B	B	Correct letter. Her last working line reads "2, 2" with the minus lost; no mark affected.

Three things about how she answered the MCQs

A calculator. Beside Q1 she wrote, word for word: "solve it in the calculator because other options doesn't make any sense." Instruction 6 on the paper says a calculator is not allowed. The answer is right and the mark stands, but in a board exam she will not have one, and $\sqrt{98} + \sqrt{8}$ is exactly the radical rule she got wrong on exam-17.

Elimination instead of reasons. On Q1 and Q8 her only reason is that the other options "doesn't make sense". Elimination is fine as a check, but on her other MCQs she wrote real working and got them right.

Check the letter against the working. Q7 (working says C, letter D) and Q9 (her own substitution gave $a = 5$) show she can reach the right answer and still lose the mark in the last step. After working an MCQ, match the result to the option before writing the letter.

SECTION B — Short Questions (10.5 / 12)

Q.2(i)(a) Evaluate $\log 72$ and name the law used at each step 2 / 3

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log(2^3 x 3^2)          ∴ log mn = log m + log n
log 2^3 + log 3^2
= 3log2 + 2log3
= 3(0.3010) + 2(0.4771)
= 0.903 + 0.9542
= 1.8572

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The factor tree for 72 and the value 1.8572 are both right. She wrote the product law beside the working as a formula, but **never named either law**, and the step from $\log 2^3 + \log 3^2$ to $3 \log 2 + 2 \log 3$ (the power law) has no law written at all. The key gives this mark only when both laws are named, so it is 2/3. This is the third Maths paper where naming the law has cost a mark.

Q.2(i)(b) Write the characteristic of $\log 0.0567$ 0.5 / 1

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log 0.0567
Char:-
  0.0567
  5.67 x 10^-2

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She set it up correctly under the heading “Char:” and reached 10^{-2} , then stopped. **The answer, -2 (or 2), is never written.** Half a mark for the method. A strict board examiner would give 0 here, because the mark is for the characteristic itself.

Q.2(ii)(a) Solve $|x - 3| + 4 = 9$ and verify both solutions 2 / 2

A first attempt ($x - 3 + 4 = 9$ with the bars dropped) is fully crossed out. The second attempt isolates $|x - 3| = 5$ first, gives $x = 8$ and $x = -2$, and **verifies both** under the heading “verification”, ending “9 = 9” twice. On exam-17 she lost half a mark for leaving out the verification. Fixed.

Q.2(ii)(b) Solve $3(x - 1) \geq 2x + 4$, stating the property in each step 2 / 2

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3x - 3 >= 2x + 4      property used : Distributive property
3x - 2x - 3 >= 2x - 2x + 4 {substrative property}
x >= +3 + 4           {additive property}
x >= 7

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Correct answer, with a property named at every step. On exam-16 she named the transitive property where the additive one was used. Fixed. One small note for revision: her book calls subtracting from both sides the **additive** property too; there is no “subtractive property” in Unit 5.

Q.2(iii) Show that A(-2, -1), B(1, 3), C(7, 11) are collinear 4 / 4

$AB = 5$, $BC = 10$, $AC = 15$, all by the distance formula with every step shown, then $AB + BC = AC$ and the conclusion in her words: “It is having collinear points.” The English is clumsy but the conclusion is there, which is exactly what was missing on exam-17 3(ii)(b). Fixed.

SECTION C — Long Questions (15 / 16)**Q.3(i)(a) Square root of $x^4 - 8x^3 + 22x^2 - 24x + 9$ by division** 3 / 4

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square root = ±(x^2 - 4x + 3)^2

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The division is right: quotient $x^2 - 4x + 3$, both subtraction rows correct, zero remainder. She even remembered the $\pm$, which she left out on exam-17. But the final line has an extra square on the bracket, so as written it says the square root equals the original polynomial. One mark off for the final answer. Her divisor column is also untidy (she never wrote $2x^2 - 4x$ or $2x^2 - 8x + 3$, and marks the remainder with an X rather than 0), but the subtraction rows show she used the right divisors, so no further deduction.

Q.3(i)(b) HCF and LCM of $x^2 - 11x + 28$ and $x^2 - 3x - 28$, with verification 4 / 4

Both factorised by splitting the middle term, $HCF = x - 7$, $LCM = (x - 4)(x + 4)(x - 7)$, and the verification carried through to $(x - 7)^2(x - 4)(x + 4)$ on both sides. Full marks.

Q.3(ii) Lamp post between P(-4, 3) and Q(6, -7) 8/8

(a) $M(1, -2)$. (b) $PM = QM = 5\sqrt{2}$, simplified correctly from $\sqrt{50}$. (c) $RM = 5\sqrt{2}$, then “R is the same distance from M as P and Q.” (d) $PQ = 10\sqrt{2}$, $PM + MQ = PQ$, and: “P, M, Q are collinear, with M b/w P and Q.”

A perfect eight-mark answer, and every part ends with a written conclusion. This was the question most exposed to her old habit of stopping before the last sentence, and she did not stop.

RE-TESTS OF PAST MISTAKES

Past mistake	Where	Here
Sum-of-cubes sign pattern	exam-15 Q13, exam-16 Q11	Fixed (Q4, and Q5 too)
Evaluating $ -8 $ as -8	exam-17 Q7	Fixed (Q6)
Dropping the minus on a lower bound	exam-17 Q8	Fixed in working , letter copied wrong (Q7)
Naming the additive property	exam-16 Q18	Fixed (Q8 and 2(ii)(b))
Solving for the letter, not the element	exam-16 Q22	Not fixed (Q9)
No verification of roots	exam-17 2(v)	Fixed (2(ii)(a))
No “hence collinear”	exam-17 3(ii)(b)	Fixed (2(iii) and 3(ii)(d))
$\pm$ missing on a square root	exam-17 3(i)(b)	$\pm$ fixed , but a stray square on the answer
Not naming the log laws	exam-16 2(ii)	Not fixed (2(i)(a))
Radical addition ($\sqrt{48} + \sqrt{27} = 75$)	exam-17 Q2	Right answer, but by calculator (Q1)

Seven of ten re-tests came back fixed. The two that did not are both about the last step: giving the letter asked for, and naming a law.

WHAT TO REVISE BEFORE THE NEXT MATHS PAPER

Item	Why
Ordered pairs: answer the letter	Wrong on exam-16 and again here. If $(2a - 1, b + 3) = (9, 5)$, the question wants a , which is 5, not the element 9.
Name the law, then use it	Write “product law” and “power law” beside the steps. Cost a mark on exam-16 and here.
Finish the last line	“Characteristic = -2 ”, and the square root is $\pm(x^2 - 4x + 3)$ with no square. 1.5 marks here.
Match the letter to the working	Q7: working C, letter D.
Count the MCQs	Q3 was skipped because her numbering slipped. Number answers by the paper, and check all ten are there.
No calculator	Practise radicals by hand: $\sqrt{98} = 7\sqrt{2}$, $\sqrt{8} = 2\sqrt{2}$.

Marking method. First pass on all 9 pages, with the mark-deciding items re-read at 300 DPI. Then an independent re-verification re-rendered every page, re-derived all ten MCQ answers from the printed paper (no key errors), mapped her answers to paper questions by their working rather than her labels, ran a shift guard (label order 4/10, content mapping 7/10), and re-marked every written part. It agreed with the first pass everywhere except 2(i)(a), which it scored 2 rather than 2.5 under the key’s own rule; that correction is taken. Q3 was confirmed as skipped, not an unscanned page: paper Q6 and Q7 run continuously across the page break, and there is no label 6 or $\log_3$ working anywhere on the nine pages.